eedi-mining-misconceptions-in-mathematics

1st Place Solution Summary

Rank	1
Team	MTH 101
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Topic ID	551402
Version	Original

Source: https://www.kaggle.com/competitions/eedi-mining-misconceptions-in-mathematics/discussion/551402
Retrieved with Kaggle CLI on 2026-10-21.

I really enjoyed working on this competition and learned a lot. Big thanks to the Kaggle community for sharing great ideas and engaging discussions!

My solution pipeline includes:

- An ensemble of retrievers: `intfloat/e5-mistral-7b-instruct`, `BAAI/bge-en-icl`, and `Qwen/Qwen2.5-14B`
- 3x Re-rankers: `Qwen/Qwen2.5-14B`, `Qwen/Qwen2.5-32B` and `Qwen/Qwen2.5-72B`

All models were fine-tuned using competition data (~1.8k examples) + synthetic data (~10k examples).

Retrievers

An ensemble of retrievers was used to identify `top 32-64 misconception candidates` for each QuestionId_Answer combination using dynamic thresholds. Specifically, I kept the top 32 retrieved candidates and included up to 32 additional candidates with similarity scores within `0.06` of the top candidate.

I selected the retrievers based on their `recall@32` scores instead of `MAP@25` -- these two metrics were often inversely correlated for my models. I skipped hard mining during training as it was not helpful for `recall@32`. One important detail was to avoid having multiple demonstrations of a given misconception in a single batch.

Rerankers

I first used the `Qwen/Qwen2.5-14B` ranker to process the retrieved candidates and narrowed them down to `top 8`. For inference, I used `vllm` with `enable_prefix_caching=True` -- which significantly improved efficiency.

Next, the `Qwen/Qwen2.5-32B` ranker further reduced the candidates to the top 5, which were then passed to the `Qwen/Qwen2.5-72B` model for final ranking.

- Pointwise ranking: I used pointwise ranking for `Qwen/Qwen2.5-14B` and `Qwen/Qwen2.5-32B`. These models processed one misconception candidate at a time in their context.
- Listwise ranking: `Qwen/Qwen2.5-72B` was listwise -- sees top 5 candidates at once in the context.

The following 4 aspects were important for performance and generalization:

- **Chain of Thought (CoT)**: I first prompted `Claude 3.5 Sonnet` to generate thoughts/rationale behind picking a wrong answer. Next, I fine-tuned the Qwen2.5 models with this data. During inference, I generated CoT from the finetuned models and provided them to the rerankers to help them with ranking. Essentially, this helps to distill reasoning from the claude model.
- **Few shot examples**: I wanted to retain/leverage in-context-learning capability of the LLMs. Hence, during training I optionally included a few demonstrations of the misconception it needs to classify. Here is an example:

```
<|im_start|>system
You are an expert  detecting grade-school level math misconceptions. Verify  the incorrect answer
stems  the provided misconception.<|im_end|>
<|im_start|>user
```

```
Misconception: Thinks there are   a centimetre

Demos  the misconception:
Question: A    metre long rope has    centimetres chopped  the end.
What  the length  the rope  remains?
Give your answer  millimetres
Answer:
Misconception Answer:

--
Subject: Length Units
Topic: Convert between m  cm
Question: To convert  metres  centimetres, you should:
Correct Answer: Multiply
Incorrect Answer: Divide
Does the misconception (Thinks there are   a centimetre) lead  the incorrect answer? (Yes/No)<|
im_end|>
<|im_start|>assistant
```

During inference I used 1-2 demos from the training set. It helped the 14b ranker the most.

- **Negative Ratio**: Showing the rankers a large number of negatives during training helped to improve performance. For each positive, I sampled 24 negatives.
- **Distillation/Pseudo Labelling**: I first fine-tuned two pointwise 72b models (`Qwen/Qwen2.5-Math-72B` and `Qwen/Qwen2.5-72B`) using competition data and ensembled them to pseudo label the synthetic examples. Competition data + pseudo labelled data were used to fine-tune 14b and 32b models.

Synthetic Data

- **Grouped synthetic data generation**: I first created clusters of related misconceptions e.g.

```
>> Assumes a   linear
>> Thinks terms  linear  are  direct proportion
>> Does  know how  find the next term  a
>> Describes position  term   term  term
>> Does  recognise that a linear  must increase  decrease  same amount
>> Uses  the first two terms  a     a term--term
>> Does  know the definition  term  term--term
>>  uses two terms  a     a term--term
>>  asked  a specific term  a  just gives the next term
>> Does    changes  a   visual patterns
```

Then asked Claude 3.5 Sonnet to generate more examples. I included 5-8 relevant MCQs from the training set during the synthetic data generation. I have included the prompts in the comments.

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Links

- Inference notebook

I will write a detailed post and publish my code + datasets soon!

Supplemental Q&A

Zhongkai Shangguan · 2024-12-13T04:19:10.080000

Congrats! May I ask what's your score without reranking (retrieval only)?

InnerVoice · 2024-12-13T04:58:25.360000

Hi Raja, Nice approach. I see 32B and 72B models are finetuned .. what is the minimum GPU RAM requirement for doing the same ..
One more question can you suggest some paper articals around listwise reranking.

Raja Biswas · 2024-12-13T07:18:51.470000

I trained almost all of my models with 2xH100. Used (Q)LoRA to finetune 72b models and LoRA for the 32b ones. Both of which, needed ~80GB VRAM for my training setup.

I have came across listwise ranking from previous Kaggle solution posts, not sure if there is a good paper on this…

Raja Biswas · 2024-12-13T07:21:26.670000

My best retriever only sub had 0.524 public LB, 0.475 private LB. I didn't try to push for MAP@25 with retrievers, mostly focused on recall.

narsil (jobs-in-data.com) · 2024-12-13T08:02:19.623000

Thanks for sharing those! I actually work in a very similar manner in most of the competitions - first intuitions are often the ones the most worth checking.

Out of curiosity, why did you use Sonnet for chain of thoughts? I always look for SOTA on https://huggingface.co/spaces/lmarena-ai/chatbot-arena-leaderboard and Sonnet is not the top there

Dipam Chakraborty · 2024-12-13T08:29:30.237000

I tried one full model tuning on a 7B model and interestingly it was worse than LoRA, maybe hparam tuning could make it better though, I only reduced the max learning rate.

SiWeiLin · 2024-12-13T09:40:19.247000

Congratulations!May I ask how you grouped the misconceptions? Was it through embedding clustering?

Raja Biswas · 2024-12-13T13:44:03.570000

@narsil Nothing specific, just because I'm more familiar with claude than other LLMs. I briefly looked into o1-mini and felt that sonnet was equal or better based on vibe test..

ktr · 2024-12-13T13:59:47.900000

Congratulations on your 1st place!
I was surprised by how carefully crafted the prompt was.
I would like to know what process was used to create the prompts in the solution. Did you create it from scratch?

Raja Biswas · 2024-12-13T14:02:34.883000

I used the metaprompt notebook to get started

Raja Biswas · 2024-12-13T14:09:47.387000

Thanks for sharing @dipamc77, I previously fine-tuned LLMs for the function calling capability -- where I observed full tuning was significantly better than LoRA. In general, I felt full-tuning should have an edge for more complex tasks e.g. reasoning and coding. Maybe need to experiment more with HPs here.

Bilzard · 2024-12-13T22:57:10.203000

Congratulation for solo first prize.

I have two question

1. According to your solution, you seem to choose point-wise reranker rather than list-wise reranker if inference time is sufficiently small. What is the merit of point-wise model than list-wise model?
2. What strategy is used to clustering misconception in generating synthetic data?

Raja Biswas · 2024-12-14T00:24:51.497000

1. LLM based listwise approach typically has a position bias, rearranging the candidates would give moderately different predictions. I thought the bias would be even more stronger if more candidates are included for ranking. One way to counter this would be having multiple inference runs with different order of candidates and average the scores (like test time augmentation (TTA))
2. I used co-occurrence stats of misconceptions in re-ranker OOF. Here is a gist notebook

Raja Biswas · 2024-12-14T00:27:41.367000

I used co-occurrence stats of misconceptions in re-ranker OOF. Here is a gist notebook